Project Test Plan



ACE

CYBER R&D Program

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Project Title:<Contextualized Threat Intelligence Prioritization for CTI Analysts>

1. Test Plan Purpose and Best Practices:

This template is structured to be clear, rigorous, and promote reproducibility. It contains placeholders '[in brackets]' that you should replace with your project's specific information. The explanatory notes in italics are for your guidance and should be removed from the final document.

[*Contextualized Threat Intelligence Prioritization for CTI Analysts*]

Test Plan

Document ID:	TP-001
Document Title:	Threat Intelligence Prioritization Test Plan
Project Title:	Contextualized Threat Intelligence Prioritization for CTI Analysts
Version #:	3.0
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Author(s):	Denzel Jefferson
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1.0 Objective

This document outlines the test plan for evaluating whether including operational environmental data from Elasticsearch into a contextualized threat intelligence prioritization system using a weighted scoring model improves CTI analyst performance.

Hypothesis: If operational environment data from Elasticsearch is incorporated into an automated MISP-based threat-prioritization system, then the time required for CTI analysts to identify high-priority threats will significantly decrease while maintaining or improving overall prioritization accuracy compared to a manual review process.

This test plan will directly measure the relationship between the independent variable, the method of threat prioritization, and the dependent variables, which include prioritization accuracy, alert fatigue statistics, and analyst response time.

2.0 Scope

2.1 In-Scope

Testing of:

- *Elasticsearch log ingestion and contextual data analysis*

- *Simulated MISP-style threat intelligence scoring using a weighted contextual model*
- *Python-based automation and prioritization scripts*

Execution of:

- *Synthetic CTI alert generation and operational log simulation*
- *Manual analyst threat prioritization workflow*
- *Automated analyst threat prioritization workflow*

Measurement of:

- *Time to identify high-priority threats*
- *Threat prioritization accuracy (High, Medium, Low)*
- *Alert fatigue statistics, including alert volume reduction and signal-to-noise ratio*

Testing on the specific environment defined in the Testbed Configuration Plan -
Virtualized testbed environment defined in the Testbed Setup and Configuration Assignment

2.2 Out-of-Scope

List items that are out of scope of your testing:

- *Vulnerability scanning of the underlying operating systems*
- *Hardware performance testing*
- *Adversary simulation, such as simulated real-world threat actors' behaviors beyond synthetic alert generation*
- *Long-term testing beyond the defined experiment test cases*

3.0 Test Environment

All testing will be conducted in the established virtual testbed environment defined in the Testbed Setup and Configuration Assignment.

3.1 Key Components

Note: Adapt this section as necessary to fit your research project by changing the column name headers and editing the table.

Hostname	Role	Key SW / Config.
vm-elastic	Log ingestion	Elasticsearch, Kibana, Filebeat

host-machine	Research workstation	VirtualBox
Vm-log-generator	Synthetic log generation	Python log generator scripts

4.0 Prerequisites / Entry Criteria

The following conditions must be met before commencing any test case execution:

1. *All virtual machines in the virtual testbed are powered on and connected to the lab network.*
2. *Elasticsearch and MISP services are tested and confirmed to be running on the virtual machines to be used.*
3. *Threat intelligence datasets imported into MISP are anonymized and contain no classified information, PII, or CUI.*
4. *Base system performance (CPU and Memory) is within normal operating limits before the test.*
5. *The researcher is prepared to document the results in the Lab notebook for analysis and documentation.*

5.0 Test Cases

This is the core execution section where each test case is a distinct experiment.

Test Case ID: *TC-001*

Title: *Baseline Log processing without automated threat prioritization*

Objective*: To measure system performance and analyst workload without automated threat prioritization.*

Step-by-Step Procedure:

1. *Start synthetic log generation to simulate normal security events*
2. *Start ingesting logs into Elasticsearch*
3. *Observe event volume, alert counts, and analyst review*
4. *Record processing time and the alert distribution*
5. *Stop Log collection*

Data to be Collected:

- *Total number of alerts generated*
- *Number of unique threat events requiring review*
- *Estimated total review time*

Expected Outcomes / Success Criteria:

- *High alert volume without prioritization*
- *Analyst workload is high and unfiltered*

Test Case ID: *TC-002*

Title*: Automated threat intelligence prioritization*

Objective*: To evaluate the effectiveness of using automated prioritization using MISP-based scoring and contextual metadata.*

Step-by-Step Procedure:

1. Enable prioritization scoring logic by using simulated MISP-based scoring logic with Elasticsearch metadata

2. Re-run synthetic log generation

3. Observe automated prioritized alert output

4. Compare alert reduction and analyst workload to the baseline (non-automated) prioritization process

5. Stop log collection

Data to be Collected:

- *Prioritization output (High / Medium / Low classification)*
- *Prioritized alert list*
- *Alert reduction percentage*
- *Estimated analyst review time (Number of High + Medium Alerts X standard average review time)*
- *Time required to identify high-priority threats*

Expected Outcomes / Success Criteria:

- *Reduced alert volume*
- *High-risk threats ranked first*
- *Improved analyst efficiency*

Test Case ID: *TC-003*

Title*: High-volume Log test*

Objective*: To evaluate system performance and prioritization accuracy under high-volume log ingestion conditions.*

Step-by-Step Procedure:

1. *Increase synthetic log generation rate*
2. *Monitor Elasticsearch ingestion performance*
3. *Measure prioritization accuracy and processing time*
4. *Stop log collection*

Data to be Collected:

- *Event ingestion*
- *System CPU and memory utilization*
- *Prioritization response delay (Measured from the moment Elasticsearch ingests a log to the moment the corresponding prioritization score is updated by the PoC (proof-of-concept))*

Expected Outcomes / Success Criteria:

- *System maintains prioritization accuracy*
- *System performance under increased load is evaluated*

6.0 Data Analysis Plan

The purpose of this section is to describe how data from TC-001 and TC-002 will be compared to prove the research hypothesis.

1. *Compare the total alert volume between baseline and automated prioritization*
2. *Compare the number of alerts requiring manual analyst review*
3. *Calculate the estimated total review workload time for both cases*
4. *Measure alert reduction percentage*
5. *Evaluate prioritization accuracy by analyzing whether high-ranked alerts are ranked above lower-ranked alerts*

The research hypotheses will be supported if:

6. *Estimated analyst review time decreases under automated prioritization*
7. *Alert volume requiring manual review is reduced*
8. *Higher-ranked alerts are consistently ranked at the top of the prioritization list*

7.0 Exit Criteria & Teardown

Upon completion of all test cases in this plan, the following steps must be performed:

9. *All temporary synthetic log generation and data ingestion information/data will be disposed of in accordance with DoD rules and university guidelines.*
10. *All collected datasets will be safely stored in a research lab storage environment.*
11. *Virtual machines will be reverted to their original blank state to restore snapshots for future experiments.*

8.0 Roles & Responsibilities

Role	Name	Responsibility
Test Executor	*Denzel Jefferson*	*Execute test cases, collect data, and analyze results*
Faculty Advisor	*Professor Julie Henderson*	*Reviewing and approving results*
DoD/ACE Mentor	*Professor William Littleton*	*Reviewing and approving the Test Plan for execution.*

9.0 References

ID #	Document Title	Location / Link
1	*Project Charter*	*https://docs.google.com/document/d/1_i_rbvBvL4vRITS5wFigVs6PpBXwuuGv/edit?usp=sharing&ouid=117579668533528162480&rtpof=true&sd=true*

2	*Project Research Hypothesis*	*https://docs.google.com/document/d/1BPuY8DRDKl7m-AqPtu47l6lpeYU-G-ig/edit?usp=sharing&ouid=117579668533528162480&rtpof=true&sd=true*
3	*Project Preliminary Literature Map*	*https://docs.google.com/document/d/12dB0K1wW8Lz-ckWcD3pjYSJM0RiZEjGU/edit?usp=sharing&ouid=117579668533528162480&rtpof=true&sd=true*
4	*Project Research Design*	*https://docs.google.com/document/d/1LFUvAn-bPUw64Z4KzE0Y26OkCua-shHW/edit?usp=sharing&ouid=117579668533528162480&rtpof=true&sd=true*
5	*Project Testbed Setup & Configuration*	*https://docs.google.com/document/d/1EEWX6e4gfT6wZG5XLXryEm5DttUqr86-/edit?usp=sharing&ouid=117579668533528162480&rtpof=true&sd=true*
6	*Project Data Acq & Synthetic Data Gen Report*	*https://docs.google.com/document/d/1m1UDbaqAVyiPx8ci85B1ydYO-OxobSZ1/edit?usp=sharing&ouid=117579668533528162480&rtpof=true&sd=true*
7	*Project Data Analysis & Synthesis*	*https://docs.google.com/document/d/1oaGWHoZh1yxBoP7VKFrXrM3d8STEvySA/edit*
8	*Project Research Notebook*	*https://docs.google.com/document/d/1y5myhJ1xNX8gqfD0A-60T0oEDEpIxhsR/edit*
9	*R&D_Final-Project-Brief*	*https://docs.google.com/presentation/d/1AyHkgj8CWdzsIMtS7bvph0BudJOsoMQ1/edit?slide=id.p8#slide=id.p8*
10	*Applied Research Project Whitepaper*	*https://docs.google.com/document/d/1wFUnlhb9HwcQMaCxdAdUVm-_tnTCuyHl/edit*

Revision History

Each time you revise your test plan after your initial draft, update this section to track all of your changes.

Ver. #	Date	Author(s)	Summary of Change
1.0	2026-02-23	Denzel Jefferson	Initial draft of test plan
2.0	2026-04-29	Denzel Jefferson	Revisions
3.0	2026-05-04	Denzel Jefferson	Final version of test plan
